Julia Ebert

Robotics Research & Development • Software Engineering • Boston, MA

juliaebert.com
julia@juliaebert.com
github.com/jtebert
linkedin.com/in/jtebert
617-949-0214

Skills

Tools	Linux • Docker • Git • ZeroMQ • Protobuf • ROS • PyTorch • Reinforcement learning • MuJoCo
Languages	C++ • Python • JavaScript • HTML/CSS
Everything else	Computer-aided design (OnShape) • 3D printing • Laser cutting

Experience

Boston, MA — 2023 –
Fleet Robotics
Autonomy Lead
- Architect and implement the software stack from the ground up (hardware choice, communication protocols, Docker/CI, and UI development), enabling Fleet's first on-ship robot demonstrations.
- Develop robust path planning (A*, RL) and execution systems for autonomous underwater navigation on complex surfaces, addressing challenges of limited sensing and harsh operating conditions.
- Design and implement robot control systems, safety mechanisms, and a robust logging system.
- Manage a team of 3 software engineers and supervise interns.
- Lead software project planning and collaborate with stakeholders to manage timelines and risks.

Brighton, CO — 2022 – 2023
Outrider
Software Engineer, Mission Planning (Remote)
- Spearheaded design and development (C++, ROS) of new multi-robot planning for Outrider's fleet of autonomous distribution yard trucks.
- Led cross-functional project teams to create new robot behaviors toward commercial product goals.
- Supported test site and customer deployments of the mission planning system.

Cambridge, MA — 2016 – 2022
Harvard University Self-Organizing Systems Research Group, Prof. Radhika Nagpal
Doctoral Researcher
- Developed collective spatial decision-making algorithms for simulated and physical robot collectives. Includes bio-inspired and Bayesian decision and movement algorithms.
- Collaborated with MIT Media Lab to create heterogeneous robot swarm for inspection on space stations, including algorithm development and hardware testing in microgravity.
- Supervised undergraduate and masters research, and taught robotics and digital fabrication courses.

Livermore, CA — Summer 2018
Lawrence Livermore National Laboratory, Dr. Michael Schneider
Computational Science Research Intern
- Programmed, refactored, and documented research codebase (Python) for space situational awareness (SSA), since used extensively by researchers at LLNL and in review for JOSS publication.
- Developed a simulator and visualization tools (Python) for orbit observation by low earth orbit satellites.

Education

Cambridge, MA — 2022
Harvard University
PhD in Computer Science
- Thesis: *Distributed Decision-making for Inspection by Autonomous Robot Collectives*
- Department of Energy Computation Science Graduate Fellow (DOE CSGF) • Siebel Scholar

London, UK — 2016
Imperial College London
Master of Research (MRes) in Bioengineering, with Distinction
- Marshall Scholar • Thesis: *Assisting Balance Recovery with a Lower Limb Exoskeleton*

Boston, MA — 2015
Northeastern University
BS in Behavioral Neuroscience, Minor in Computer Science
- Goldwater Scholar • German Academic Exchange Service (DAAD) Scholar • summa cum laude

Select Publications

J Meyers, M Schneider, **J Ebert** et al. (2025) SSAPy – Space Situational Awareness for Python. *JOSS.*

J Ebert et al. (2022) A Hybrid PSO Algorithm for Multi-robot Target Search and Decision Awareness. *IROS 2022.*

J Ebert et al. (2020) Bayes Bots: Collective Bayesian Decision-Making in Decentralized Robot Swarms. *ICRA 2020.*

J Ebert et al. (2018) Multi-feature collective decision making in robot swarms. *AAMAS 2018.*

S Bazzi, **J Ebert** et al. (2018) Stability and predictability in human control of complex objects. *Chaos*, 28, 10. .